HR Analytics Using SQL

End-to-End Data Cleaning and Business Analysis Project

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Tools: MySQL, SQL

Project Type: Portfolio Project

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Project Overview

This project demonstrates an end-to-end HR analytics workflow using MySQL. The dataset contains employee and training information. Raw data was cleaned, validated, standardized, and analyzed to generate business insights for workforce planning, employee performance evaluation, training effectiveness, and executive reporting.

Business Objective

Objective
Improve HR data quality
Remove duplicate records
Standardize inconsistent values
Analyze employee and training data
Generate HR business insights
Support data-driven decision making

Dataset Information

Item	Details
Domain	Human Resources
Database	MySQL

Records	3,000
Columns	28
Dataset Type	Structured Employee Data

Project Workflow

Raw Dataset → Import into MySQL → Data Cleaning → Data Validation → Duplicate Detection → Duplicate Removal → Business Analysis → Executive Insights

Data Cleaning Process

Step	Description
Data Type Conversion	Converted text dates into DATE format
Text Standardization	Removed leading and trailing spaces
Missing Value Check	Validated NULL values
Duplicate Detection	Identified duplicates using ROW_NUMBER()
Duplicate Removal	Removed duplicate records while preserving one copy
Validation	Verified dataset before analysis

Data Validation

After cleaning, the dataset was validated to ensure consistent date formats, standardized text values, and the absence of duplicate records before business analysis.

Duplicate Detection & Removal

Duplicate records were identified using the ROW_NUMBER() window function. A temporary table with a row number column was created, duplicate rows were removed, and the helper column was dropped after verification.

SQL Techniques Used

Category	Techniques
Data Cleaning	TRIM(), UPDATE, ALTER TABLE
Aggregation	COUNT(), AVG(), SUM()
Conditional Logic	CASE
Window Functions	ROW_NUMBER(), DENSE_RANK()
CTE	Common Table Expressions
Filtering	WHERE, HAVING
Grouping	GROUP BY

Business Analysis

The project answers 15 HR business questions covering workforce distribution, department performance, employee satisfaction, engagement, attrition, training effectiveness, employee experience, and an executive HR dashboard.

Key Business Insights

• Total Employees: 3,000

• Overall Attrition Rate: 12.90%

• Admin Offices achieved the highest average performance rating.

• Sales recorded the highest employee satisfaction.

• Executive Office achieved the highest Employee Experience Score.

• Communication Skills is the most popular training program.

Conclusion

This project demonstrates a complete SQL analytics workflow from raw HR data to business insights. It showcases practical data cleaning, validation, duplicate removal, advanced SQL techniques, and business-focused reporting suitable for a Data Analyst portfolio.